Withdrawn NIST Technical Series Publication

Warning Notice

The attached publication has been withdrawn (archived), and is provided solely for historical purposes. It may have been superseded by another publication (indicated below).

Withdrawn Publication	
Series/Number	Federal Information Processing Standards Publication 46
Title	Data Encryption Standard
Publication Date(s)	January 15, 1977
Withdrawal Date	January 22, 1988
Withdrawal Note	FIPS 46 was superseded and reaffirmed by FIPS 46-1 (January 22, 1988)

Superseding Publication(s) *(if applicable)*

The attached publication has been **superseded by** the following publication(s):

Series/Number	Federal Information Processing Standards Publication 46-1
Title	Data Encryption Standard
Author(s)	National Bureau of Standards
Publication Date(s)	January 22, 1988
URL/DOI	https://csrc.nist.gov/publications/detail/fips/46/1/archive/1988-01-22

Additional Information *(if applicable)*	
Contact	Computer Security Division (Information Technology Laboratory)
Latest revision of the attached publication	
Related Information	
Withdrawal Announcement Link	

Date updated: January 8, 2020

National Institute of Standards and Technology
U.S. Department of Commerce



FIPS PUB 46

FEDERAL INFORMATION PROCESSING STANDARDS PUBLICATION
1977 JANUARY 15

U.S. DEPARTMENT OF COMMERCE / National Bureau of Standards



DATA ENCRYPTION STANDARD

CATEGORY: ADP OPERATIONS
SUBCATEGORY: COMPUTER SECURITY

U.S. DEPARTMENT OF COMMERCE • Elliot L. Richardson, *Secretary*

Edward O. Vetter, *Under Secretary*

Dr. Betsy Ancker-Johnson, *Assistant Secretary for Science and Technology*

NATIONAL BUREAU OF STANDARDS • Ernest Ambler, *Acting Director*

Foreword

The Federal Information Processing Standards Publication Series of the National Bureau of Standards is the official publication relating to standards adopted and promulgated under the provisions of Public Law 89-306 (Brooks Bill) and under Part 6 of Title 15, Code of Federal Regulations. These legislative and executive mandates have given the Secretary of Commerce important responsibilities for improving the utilization and management of computers and automatic data processing systems in the Federal Government. To carry out the Secretary's responsibilities, the NBS, through its Institute for Computer Sciences and Technology, provides leadership, technical guidance, and coordination of government efforts in the development of technical guidelines and standards in these areas.

The series is used to announce Federal Information Processing Standards, and to provide standards information of general interest and an index of relevant standards publications and specifications. Publications that announce adoption of standards provide the necessary policy, administrative, and guidance information for effective standards implementation and use. The technical specifications of the standard are usually attached to the publication, otherwise a reference source is cited.

Comments covering Federal Information Processing Standards and Publications are welcomed, and should be addressed to the Associate Director for ADP Standards, Institute for Computer Sciences and Technology, National Bureau of Standards, Washington, D.C. 20234. Such comments will be either considered by NBS or forwarded to the responsible activity as appropriate.

ERNEST AMBLER, *Acting Director*

Abstract

The selective application of technological and related procedural safeguards is an important responsibility of every Federal organization in providing adequate security to its ADP systems. This publication provides a standard to be used by Federal organizations when these organizations specify that cryptographic protection is to be used for sensitive or valuable computer data. Protection of computer data during transmission between electronic components or while in storage may be necessary to maintain the confidentiality and integrity of the information represented by that data. The standard specifies an encryption algorithm which is to be implemented in an electronic device for use in Federal ADP systems and networks. The algorithm uniquely defines the mathematical steps required to transform computer data into a cryptographic cipher. It also specifies the steps required to transform the cipher back to its original form. A device performing this algorithm may be used in many applications areas where cryptographic data protection is needed. Within the context of a total security program comprising physical security procedures, good information management practices and computer system/network access controls, the Data Encryption Standard is being made available for use by Federal agencies.

Key Words: ADP security; computer security; encryption; Federal Information Processing Standard.

Nat. Bur. Stand. (U.S.), Fed. Info. Process. Stand. Publ. (FIPS PUB) 46, 17 pages (1977)
CODEN: FIPPAT

For sale by the National Technical Information Service, U.S. Department of Commerce, Springfield, Virginia 22161

Federal Information Processing Standards Publication 46

1977 January 15

ANNOUNCING THE

DATA ENCRYPTION STANDARD

Federal Information Processing Standards are issued by the National Bureau of Standards pursuant to the Federal Property and Administrative Services Act of 1949, as amended, Public Law 89-306 (79 Stat 1127), Executive Order 11717 (38 FR 12315, dated May 11, 1973), and Part 6 of Title 15 Code of Federal Regulations (CFR).

Name of Standard: Data Encryption Standard (DES).

Category of Standard: Operations, Computer Security.

Explanation: The Data Encryption Standard (DES) specifies an algorithm to be implemented in electronic hardware devices and used for the cryptographic protection of computer data. This publication provides a complete description of a mathematical algorithm for encrypting (enciphering) and decrypting (deciphering) binary coded information. Encrypting data converts it to an unintelligible form called cipher. Decrypting cipher converts the data back to its original form. The algorithm described in this standard specifies both enciphering and deciphering operations which are based on a binary number called a key. The key consists of 64 binary digits ("0"s or "1"s) of which 56 bits are used directly by the algorithm and 8 bits are used for error detection.

Binary coded data may be cryptographically protected using the DES algorithm in conjunction with a key. The key is generated in such a way that each of the 56 bits used directly by the algorithm are random and the 8 error detecting bits are set to make the parity of each 8-bit byte of the key odd, i.e., there is an odd number of "1"s in each 8-bit byte. Each member of a group of authorized users of encrypted computer data must have the key that was used to encipher the data in order to use it. This key, held by each member in common, is used to decipher the data received in cipher form from other members of the group. The encryption algorithm specified in this standard is commonly known among those using the standard. The unique key chosen for use in a particular application makes the results of encrypting data using the algorithm unique. Selection of a different key causes the cipher that is produced for any given set of inputs to be different. The cryptographic security of the data depends on the security provided for the key used to encipher and decipher the data.

Data can be recovered from cipher only by using exactly the same key used to encipher it. Unauthorized recipients of the cipher who know the algorithm but do not have the correct key cannot derive the original data algorithmically. However, anyone who does have the key and the algorithm can easily decipher the cipher and obtain the original data. A standard algorithm based on a secure key thus provides a basis for exchanging encrypted computer data by issuing the key used to encipher it to those authorized to have the data. Additional FIPS guidelines for implementing and using the DES are being developed and will be published by NBS.

Approving Authority: Secretary of Commerce.

Maintenance Agency: Institute for Computer Sciences and Technology, National Bureau of Standards.

Applicability: This standard will be used by Federal departments and agencies for the cryptographic protection of computer data when the following conditions apply:

1. An authorized official or manager responsible for data security or the security of any computer system decides that cryptographic protection is required; and
2. The data is not classified according to the National Security Act of 1947, as amended, or the Atomic Energy Act of 1954, as amended.

However, Federal agencies or departments which use cryptographic devices for protecting data classified according to either of these acts can use those devices for protecting unclassified data in lieu of the standard.

In addition, this standard may be adopted and used by non-Federal Government organizations. Such use is encouraged when it provides the desired security for commercial and private organizations.

Data that is considered sensitive by the responsible authority, data that has a high value, or data that represents a high value should be cryptographically protected if it is vulnerable to unauthorized disclosure or undetected modification during transmission or while in storage. A risk analysis should be performed under the direction of a responsible authority to determine potential threats. FIPS PUB 31 (Guidelines for Automatic Data Processing Physical Security and Risk Management) and FIPS PUB 41 (Computer Security Guidelines for Implementing the Privacy Act of 1974) provide guidance for making such an analysis. The costs of providing cryptographic protection using this standard as well as alternative methods of providing this protection and their respective costs should be projected. A responsible authority then should make a decision, based on these analyses, whether or not to use cryptographic protection and this standard.

Applications: Data encryption (cryptography) may be utilized in various applications and in various environments. The specific utilization of encryption and the implementation of the DES will be based on many factors particular to the computer system and its associated components. In general, cryptography is used to protect data while it is being communicated between two points or while it is stored in a medium vulnerable to physical theft. Communication security provides protection to data by enciphering it at the transmitting point and deciphering it at the receiving point. File security provides protection to data by enciphering it when it is recorded on a storage medium and deciphering it when it is read back from the storage medium. In the first case, the key must be available at the transmitter and receiver simultaneously during communication. In the second case, the key must be maintained and accessible for the duration of the storage period.

Hardware Implementation: The algorithm specified in this standard is to be implemented in computer or related data communication devices using hardware (not software) technology. The specific implementation may depend on several factors such as the application, the environment, the technology used, etc. Implementations which comply with this standard include Large Scale Integration (LSI) "chips" in individual electronic packages, devices built from Medium Scale Integration (MSI) electronic components, or other electronic devices dedicated to performing the operations of the algorithm. Micro-processors using Read Only Memory (ROM) or micro-programmed devices using microcode for hardware level control instructions are examples of the latter. Hardware implementations of the algorithm which are tested and validated by NBS will be considered as complying with the standard. Procedures for testing and validating equipment for conformance with this standard are available from the Systems and Software Division, National Bureau of Standards, Washington, D.C. 20234. Software implementations in general purpose computers are not in compliance with this standard. Information regarding devices which have been tested and validated will be made available to all FIPS points of contact.

Export Control: Cryptographic devices and technical data regarding them are subject to Federal Government export controls as specified in Title 22, Code of Federal Regulations, Parts 121 through 128. Cryptographic devices implementing this standard and technical data regarding them must comply with these Federal regulations.

Patents: Crytographic devices implementing this standard may be covered by U.S. and foreign patents issued to the International Business Machines Corporation. However, IBM has granted nonexclusive, royalty-free licenses under the patents to make, use and sell apparatus which complies with the standard. The terms, conditions and scope of the licenses are set out in notices published in the May 13, 1975 and August 31, 1976 issues of the Official Gazette of the United States Patent and Trademark Office (934 O. G. 452 and 949 O. G. 1717).

Alternative Modes of Using the DES: The "Guidelines for Implementing and Using the Data Encryption Standard" describe two different modes for using the algorithm described in this standard. Blocks of data containing 64 bits may be directly entered into the device where 64-bit cipher blocks are generated under control of the key. This is called the electronic code book mode. Alternatively, the device may be used as a binary stream generator to produce statistically random binary bits which are then combined with the clear (unencrypted) data (1-64 bits) using an "exclusive-or" logic operation. In order to assure that the enciphering device and the deciphering device are synchronized, their inputs are always set to the previous 64 bits of cipher that were transmitted or received. This second mode of using the encryption algorithm is called the cipher feedback (CFB) mode. The electronic codebook mode generates blocks of 64 cipher bits. The cipher feedback mode generates cipher having the same number of bits as the plain text. Each block of cipher is independent of all others when the electronic codebook mode is used while each byte (group of bits) of cipher depends on the previous 64 cipher bits when the cipher feedback mode is used. The modes of operation briefly described here are further explained in the FIPS "Guidelines for Implementing and Using the Data Encryption Standard."

Implementation of this standard: This standard becomes effective six months after the publication date of this FIPS PUB. It applies to all Federal ADP systems and associated telecommunications networks under development as well as to installed systems when it is determined that cryptographic protection is required. Each Federal department or agency will issue internal directives for the use of this standard by their operating units based on their data security requirement determinations.

NBS will provide assistance to Federal organizations by developing and issuing additional technical guidelines on computer security and by providing technical assistance in using data encryption. A data encryption testbed has been established within NBS for use in providing this technical assistance. The National Security Agency assists Federal departments and agencies in communications security and in determining specific security requirements. Instructions and regulations for procuring data processing equipment utilizing this standard will be provided by the General Services Administration.

Specifications: Federal Information Processing Standard (FIPS 46) Data Encryption Standard (DES) (affixed).

Cross Index:

a. FIPS PUB 31, "Guidelines to ADP Physical Security and Risk Management"

b. FIPS PUB 39, "Glossary for Computer Systems Security"

c. FIPS PUB 41, "Computer Security Guidelines for Implementing the Privacy Act of 1974"

d. FIPS PUB—, "Guidelines for Implementing and Using the Data Encryption Standard" (to be published)

e. Other FIPS and Federal Standards are applicable to the implementation and use of this standard. In particular, the American Standard Code for Information Interchange (FIPS PUB 1)

and other related data storage media or data communications standards should be used in conjunction with this standard. A list of currently approved FIPS may be obtained from the Office of ADP Standards Management, Institute for Computer Sciences and Technology, National Bureau of Standards, Washington, D.C. 20234.

Qualifications: The cryptographic algorithm specified in this standard transforms a 64-bit binary value into a unique 64-bit binary value based on a 56-bit variable. If the complete 64-bit input is used (i.e., none of the input bits should be predetermined from block to block) and if the 56-bit variable is randomly chosen, no technique other than trying all possible keys using known input and output for the DES will guarantee finding the chosen key. As there are over 70,000,000,000,000,000 (seventy quadrillion) possible keys of 56 bits, the feasibility of deriving a particular key in this way is extremely unlikely in typical threat environments. Moreover, if the key is changed frequently, the risk of this event is greatly diminished. However, users should be aware that it is theoretically possible to derive the key in fewer trials (with a correspondingly lower probability of success depending on the number of keys tried) and should be cautioned to change the key as often as practical. Users must change the key and provide it a high level of protection in order to minimize the potential risks of its unauthorized computation or acquisition. The feasibility of computing the correct key may change with advances in technology. A more complete description of the strength of this algorithm against various threats will be contained in the Guidelines for Implementing and Using the DES.

When correctly implemented and properly used, this standard will provide a high level of cryptographic protection to computer data. NBS, supported by the technical assistance of Government agencies responsible for communication security, has determined that the algorithm specified in this standard will provide a high level of protection for a time period beyond the normal life cycle of its associated ADP equipment. The protection provided by this algorithm against potential new threats will be reviewed within five years to assess its adequacy. In addition, both the standard and possible threats reducing the security provided through the use of this standard will undergo continual review by NBS and other cognizant Federal organizations. The new technology available at that time will be evaluated to determine its impact on the standard. In addition, the awareness of any breakthrough in technology or any mathematical weakness of the algorithm will cause NBS to reevaluate this standard and provide necessary revisions.

Comments: Comments and suggestions regarding this standard and its use are welcomed and should be addressed to the Associate Director for ADP Standards, Institute for Computer Sciences and Technology, National Bureau of Standards, Washington, D.C. 20234.

Waiver Procedure: The head of a Federal agency may waive the provisions of this FIPS PUB after the conditions and justifications for the waiver have been coordinated with the National Bureau of Standards. A waiver is necessary if cryptographic devices performing an algorithm other than that which is specified in this standard are to be used by a Federal agency for data subject to cryptographic protection under this standard. No waiver is necessary if classified communications security equipment is to be used. Software implementations of this algorithm for operational use in general purpose computer systems do not comply with this standard and each such implementation must also receive a waiver. Implementation of the algorithm in software for testing or evaluation does not require waiver approval. Implementation of other special purpose cryptographic algorithms in software for limited use within a computer system (e.g., encrypting password files) or implementations of cryptographic algorithms in software which were being utilized in computer systems before the effective date of this standard do not require a waiver. However, these limited uses should be converted to the use of this standard when the system or equipment involved is upgraded or redesigned to include general cryptographic protection of computer data. Letters describing the nature of and reasons for the waiver should be addressed to the Associate Director for ADP Standards as previously noted.

Sixty days should be allowed for review and response by NBS. The waiver shall not be approved until a response from NBS is received; however, the final decision for granting the waiver is the responsibility of the head of the particular agency involved.

Where to Obtain Copies of the Standard:

Copies of this publication are for sale by the National Technical Information Service, U. S. Department of Commerce, 5285 Port Royal Road, Springfield, Virginia 22161. Order by FIPS PUB number and title. Prices are published by NTIS in current catalogs and other issuances. Payment may be made by check, money order, deposit account or charged to a credit card accepted by NTIS.

Blank



Federal Information Processing Standards Publication 46

1977 January 15

SPECIFICATIONS FOR THE

DATA ENCRYPTION STANDARD



The Data Encryption Standard (DES) shall consist of the following Data Encryption Algorithm to be implemented in special purpose electronic devices. These devices shall be designed in such a way that they may be used in a computer system or network to provide cryptographic protection to binary coded data. The method of implementation will depend on the application and environment. The devices shall be implemented in such a way that they may be tested and validated as accurately performing the transformations specified in the following algorithm.

DATA ENCRYPTION ALGORITHM

Introduction

The algorithm is designed to encipher and decipher blocks of data consisting of 64 bits under control of a 64-bit key. Deciphering must be accomplished by using the same key as for enciphering, but with the schedule of addressing the key bits altered so that the deciphering process is the reverse of the enciphering process. A block to be enciphered is subjected to an initial permutation IP, then to a complex key-dependent computation and finally to a permutation which is the inverse of the initial permutation IP^{-1}. The key-dependent computation can be simply defined in terms of a function f, called the cipher function, and a function KS, called the key schedule. A description of the computation is given first, along with details as to how the algorithm is used for encipherment. Next, the use of the algorithm for decipherment is described. Finally, a definition of the cipher function f is given in terms of primitive functions which are called the selection functions S_i and the permutation function P. S_i, P and KS of the algorithm are contained in the Appendix.

The following notation is convenient: Given two blocks L and R of bits, LR denotes the block consisting of the bits of L followed by the bits of R. Since concatenation is associative $B_1B_2 \ldots B_8$, for example, denotes the block consisting of the bits of B_1 followed by the bits of $B_2 \ldots$ followed by the bits of B_8.

Enciphering

A sketch of the enciphering computation is given in figure 1.

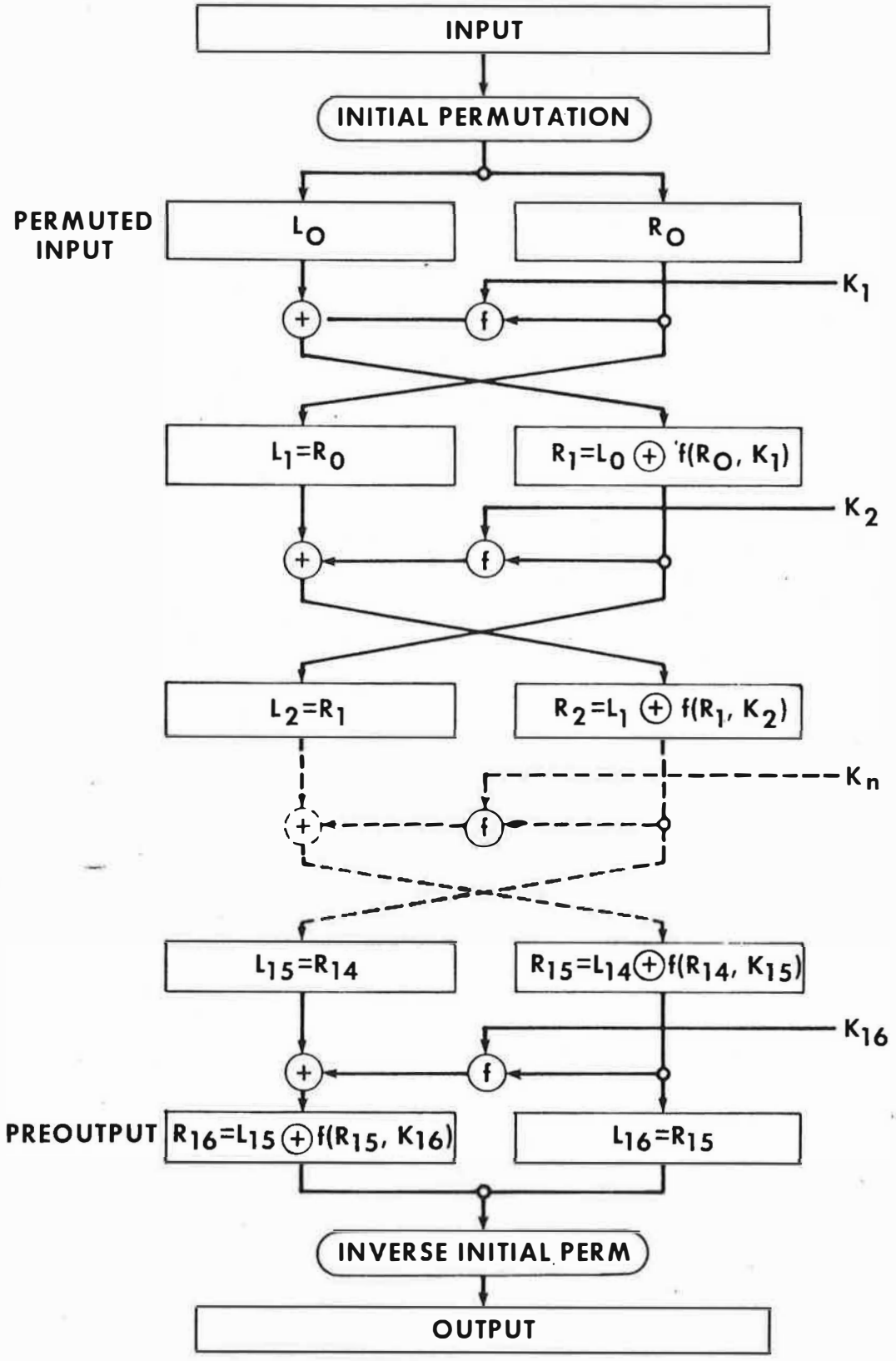


FIGURE 1. *Enciphering computation.*

The 64 bits of the input block to be enciphered are first subjected to the following permutation, called the initial permutation *IP*:

IP

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

That is the permuted input has bit 58 of the input as its first bit, bit 50 as its second bit, and so on with bit 7 as its last bit. The permuted input block is then the input to a complex key-dependent computation described below. The output of that computation, called the preoutput, is then subjected to the following permutation which is the inverse of the initial permutation:

IP^{-1}

40	8	48	16	56	24	64	32
39	7	47	15	55	23	63	31
38	6	46	14	54	22	62	30
37	5	45	13	53	21	61	29
36	4	44	12	52	20	60	28
35	3	43	11	51	19	59	27
34	2	42	10	50	18	58	26
33	1	41	9	49	17	57	25

That is, the output of the algorithm has bit 40 of the preoutput block as its first bit, bit 8 as its second bit, and so on, until bit 25 of the preoutput block is the last bit of the output.

The computation which uses the permuted input block as its input to produce the preoutput block consists, but for a final interchange of blocks, of 16 iterations of a calculation that is described below in terms of the cipher function f which operates on two blocks, one of 32 bits and one of 48 bits, and produces a block of 32 bits.

Let the 64 bits of the input block to an iteration consist of a 32 bit block L followed by a 32 bit block R. Using the notation defined in the introduction, the input block is then LR.

Let K be a block of 48 bits chosen from the 64-bit key. Then the output $L'R'$ of an iteration with input LR is defined by:

(1)
$$L' = R$$
$$R' = L \oplus f(R,K)$$

where $\oplus$ denotes bit-by-bit addition modulo 2.

As remarked before, the input of the first iteration of the calculation is the permuted input block. If $L'R'$ is the output of the 16th iteration then $R'L'$ is the preoutput block. At each iteration a different block K of key bits is chosen from the 64-bit key designated by KEY.

With more notation we can describe the iterations of the computation in more detail. Let KS be a function which takes an integer n in the range from 1 to 16 and a 64-bit block KEY as input and yields as output a 48-bit block K_n which is a permuted selection of bits from KEY. That is

$$K_n = KS(n, KEY) \tag{2}$$

with K_n determined by the bits in 48 distinct bit positions of KEY. KS is called the key schedule because the block K used in the n'th iteration of (1) is the block K_n determined by (2).

As before, let the permuted input block be LR. Finally, let L_0 and R_0 be respectively L and R and let L_n and R_n be respectively L' and R' of (1) when L and R are respectively L_{n-1} and R_{n-1} and K is K_n; that is, when n is in the range from 1 to 16,

$$\begin{aligned} L_n &= R_{n-1} \\ R_n &= L_{n-1} \oplus f(R_{n-1}, K_n) \end{aligned} \tag{3}$$

The preoutput block is then $R_{16}L_{16}$.

The key schedule KS of the algorithm is described in detail in the Appendix. The key schedule produces the 16 K_n which are required for the algorithm.

Deciphering

The permutation IP^{-1} applied to the preoutput block is the inverse of the initial permutation IP applied to the input. Further, from (1) it follows that:

$$\begin{aligned} R &= L' \\ L &= R' \oplus f(L', K) \end{aligned} \tag{4}$$

Consequently, to ***decipher*** it is only necessary to apply the ***very same algorithm to an enciphered message block,*** taking care that at each iteration of the computation ***the same block of key bits*** K ***is used*** during decipherment as was used during the encipherment of the block. Using the notation of the previous section, this can be expressed by the equations:

$$\begin{aligned} R_{n-1} &= L_n \\ L_{n-1} &= R_n \oplus f(L_n, K_n) \end{aligned} \tag{5}$$

where now $R_{16}L_{16}$ is the permuted input block for the deciphering calculation and L_0R_0 is the preoutput block. That is, for the decipherment calculation with $R_{16}L_{16}$ as the permuted input, K_{16} is used in the first iteration, K_{15} in the second, and so on, with K_1 used in the 16th iteration.

The Cipher Function f

A sketch of the calculation of $f(R, K)$ is given in figure 2.

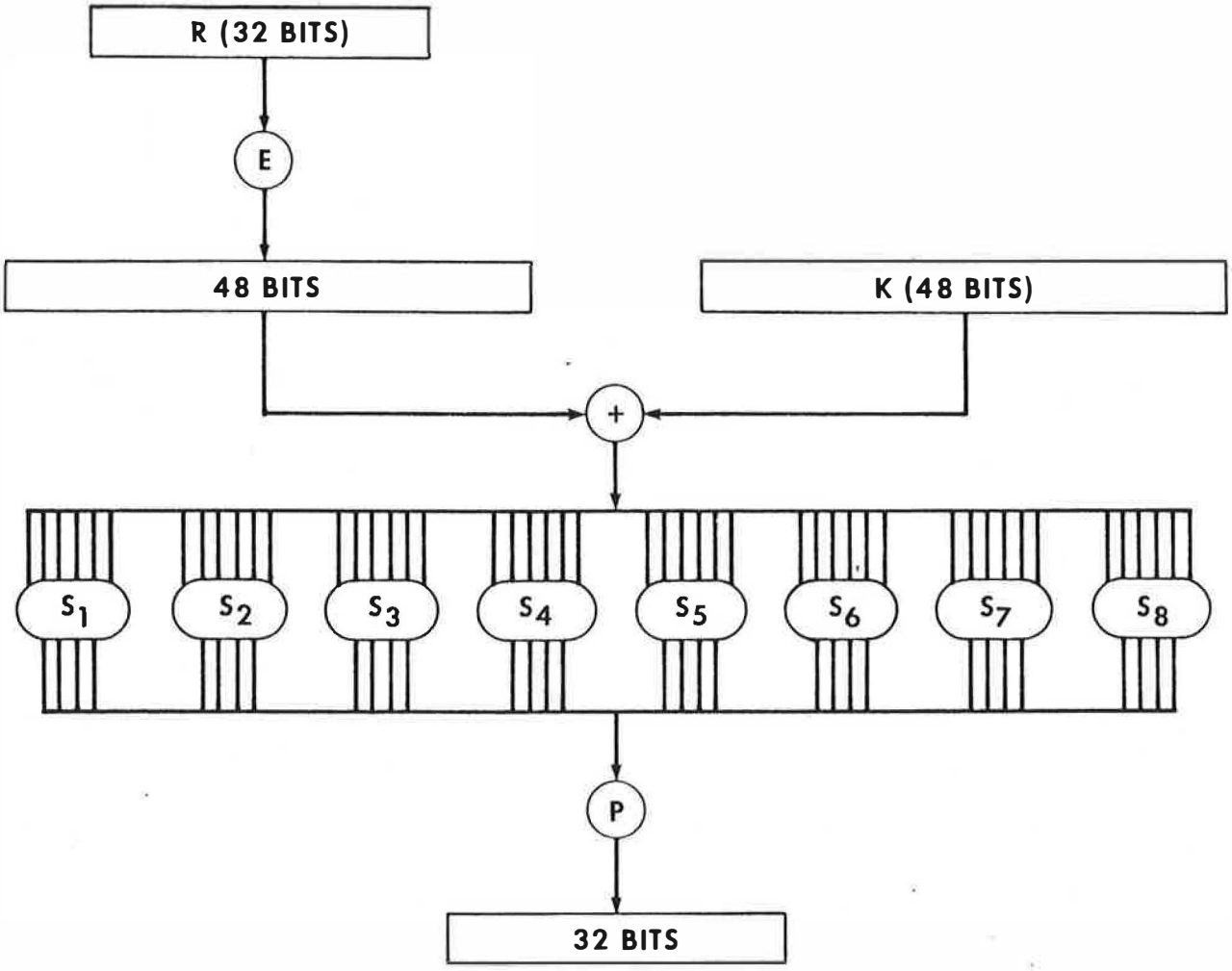


FIGURE 2. *Calculation of* f (R, K).

Let E denote a function which takes a block of 32 bits as input and yields a block of 48 bits as output. Let E be such that the 48 bits of its output, written as 8 blocks of 6 bits each, are obtained by selecting the bits in its inputs in order according to the following table:

E BIT-SELECTION TABLE

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

Thus the first three bits of $E(R)$ are the bits in positions 32, 1 and 2 of R while the last 2 bits of $E(R)$ are the bits in positions 32 and 1.

Each of the unique selection functions $S_1, S_2, \ldots, S_8$, takes a 6-bit block as input and yields a 4-bit block as output and is illustrated by using a table containing the recommended S_1:

S_1

Column Number

Row No.	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
1	0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
2	4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
3	15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

If S_1 is the function defined in this table and B is a block of 6 bits, then $S_1(B)$ is determined as follows: The first and last bits of B represent in base 2 a number in the range 0 to 3. Let that number be i. The middle 4 bits of B represent in base 2 a number in the range 0 to 15. Let that number be j. Look up in the table the number in the i'th row and j'th column. It is a number in the range 0 to 15 and is uniquely represented by a 4 bit block. That block is the output $S_1(B)$ of S_1 for the input B. For example, for input 011011 the row is 01, that is row 1, and the column is determined by 1101, that is column 13. In row 1 column 13 appears 5 so that the output is 0101. Selection functions $S_1, S_2, \ldots, S_8$ of the algorithm appear in the Appendix.

The permutation function P yields a 32-bit output from a 32-bit input by permuting the bits of the input block. Such a function is defined by the following table:

P

16	7	20	21
29	12	28	17
1	15	23	26
5	18	31	10
2	8	24	14
32	27	3	9
19	13	30	6
22	11	4	25

The output $P(L)$ for the function P defined by this table is obtained from the input L by taking the 16th bit of L as the first bit of $P(L)$, the 7th bit as the second bit of $P(L)$, and so on until the 25th bit of L is taken as the 32nd bit of $P(L)$. The permutation function P of the algorithm is repeated in the Appendix.

Now let $S_1, \ldots, S_8$ be eight distinct selection functions, let P be the permutation function and let E be the function defined above.

To define $f(R, K)$ we first define $B_1, \ldots, B_8$ to be blocks of 6 bits each for which

(6) $$B_1 B_2 \ldots B_8 = K \oplus E(R)$$

The block $f(R, K)$ is then defined to be

(7) $$P(S_1(B_1)S_2(B_2) \ldots S_8(B_8))$$

Thus $K \oplus E(R)$ is first divided into the 8 blocks as indicated in (6). Then each B_i is taken as an input to S_i and the 8 blocks $S_1(B_1)$, $S_2(B_2)$, . . ., $S_8(B_8)$ of 4 bits each are consolidated into a single block of 32 bits which forms the input to P. The output (7) is then the output of the function f for the inputs R and K.

Blank

APPENDIX

PRIMITIVE FUNCTIONS FOR THE DATA ENCRYPTION ALGORITHM

The choice of the primitive functions KS, S_1, . . ., S_8 and P is critical to the strength of an encipherment resulting from the algorithm. Specified below is the recommended set of functions, describing S_1, . . ., S_8 and P in the same way they are described in the algorithm. For the interpretation of the tables describing these functions, see the discussion in the body of the algorithm.

The primitive functions S_1, . . ., S_8, are:

S_1

14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

S_2

15	1	8	14	6	11	3	4	9	7	2	13	12	0	5	10
3	13	4	7	15	2	8	14	12	0	1	10	6	9	11	5
0	14	7	11	10	4	13	1	5	8	12	6	9	3	2	15
13	8	10	1	3	15	4	2	11	6	7	12	0	5	14	9

S_3

10	0	9	14	6	3	15	5	1	13	12	7	11	4	2	8
13	7	0	9	3	4	6	10	2	8	5	14	12	11	15	1
13	6	4	9	8	15	3	0	11	1	2	12	5	10	14	7
1	10	13	0	6	9	8	7	4	15	14	3	11	5	2	12

S_4

7	13	14	3	0	6	9	10	1	2	8	5	11	12	4	15
13	8	11	5	6	15	0	3	4	7	2	12	1	10	14	9
10	6	9	0	12	11	7	13	15	1	3	14	5	2	8	4
3	15	0	6	10	1	13	8	9	4	5	11	12	7	2	14

S_5

2	12	4	1	7	10	11	6	8	5	3	15	13	0	14	9
14	11	2	12	4	7	13	1	5	0	15	10	3	9	8	6
4	2	1	11	10	13	7	8	15	9	12	5	6	3	0	14
11	8	12	7	1	14	2	13	6	15	0	9	10	4	5	3

S_6

12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

S_7

4	11	2	14	15	0	8	13	3	12	9	7	5	10	6	1
13	0	11	7	4	9	1	10	14	3	5	12	2	15	8	6
1	4	11	13	12	3	7	14	10	15	6	8	0	5	9	2
6	11	13	8	1	4	10	7	9	5	0	15	14	2	3	12

S_8

13	2	8	4	6	15	11	1	10	9	3	14	5	0	12	7
1	15	13	8	10	3	7	4	12	5	6	11	0	14	9	2
7	11	4	1	9	12	14	2	0	6	10	13	15	3	5	8
2	1	14	7	4	10	8	13	15	12	9	0	3	5	6	11

The primitive function P is:

16	7	20	21
29	12	28	17
1	15	23	26
5	18	31	10
2	8	24	14
32	27	3	9
19	13	30	6
22	11	4	25

Recall that K_n, for $1 \leq n \leq 16$, is the block of 48 bits in (2) of the algorithm. Hence, to describe *KS*, it is sufficient to describe the calculation of K_n from *KEY* for $n = 1, 2, \ldots, 16$. That calculation is illustrated in figure 3. To complete the definition of *KS* it is therefore sufficient to describe the two permuted choices, as well as the schedule of left shifts. One bit in each 8-bit byte of the *KEY* may be utilized for error detection in key generation, distribution and storage. Bits 8, 16, . . ., 64 are for use in assuring that each byte is of odd parity.

Permuted choice 1 is determined by the following table:

PC–1

57	49	41	33	25	17	9
1	58	50	42	34	26	18
10	2	59	51	43	35	27
19	11	3	60	52	44	36
63	55	47	39	31	23	15
7	62	54	46	38	30	22
14	6	61	53	45	37	29
21	13	5	28	20	12	4

The table has been divided into two parts, with the first part determining how the bits of C_0 are chosen, and the second part determining how the bits of D_0 are chosen. The bits of *KEY* are numbered 1 through 64. The bits of C_0 are respectively bits 57, 49, 41, . . ., 44 and 36 of *KEY*, with the bits of D_0 being bits 63, 55, 47, . . ., 12 and 4 of *KEY*.

With C_0 and D_0 defined, we now define how the blocks C_n and D_n are obtained from the blocks C_{n-1} and D_{n-1}, respectively, for $n = 1, 2, \ldots, 16$. That is accomplished by adhering to the following schedule of left shifts of the individual blocks:

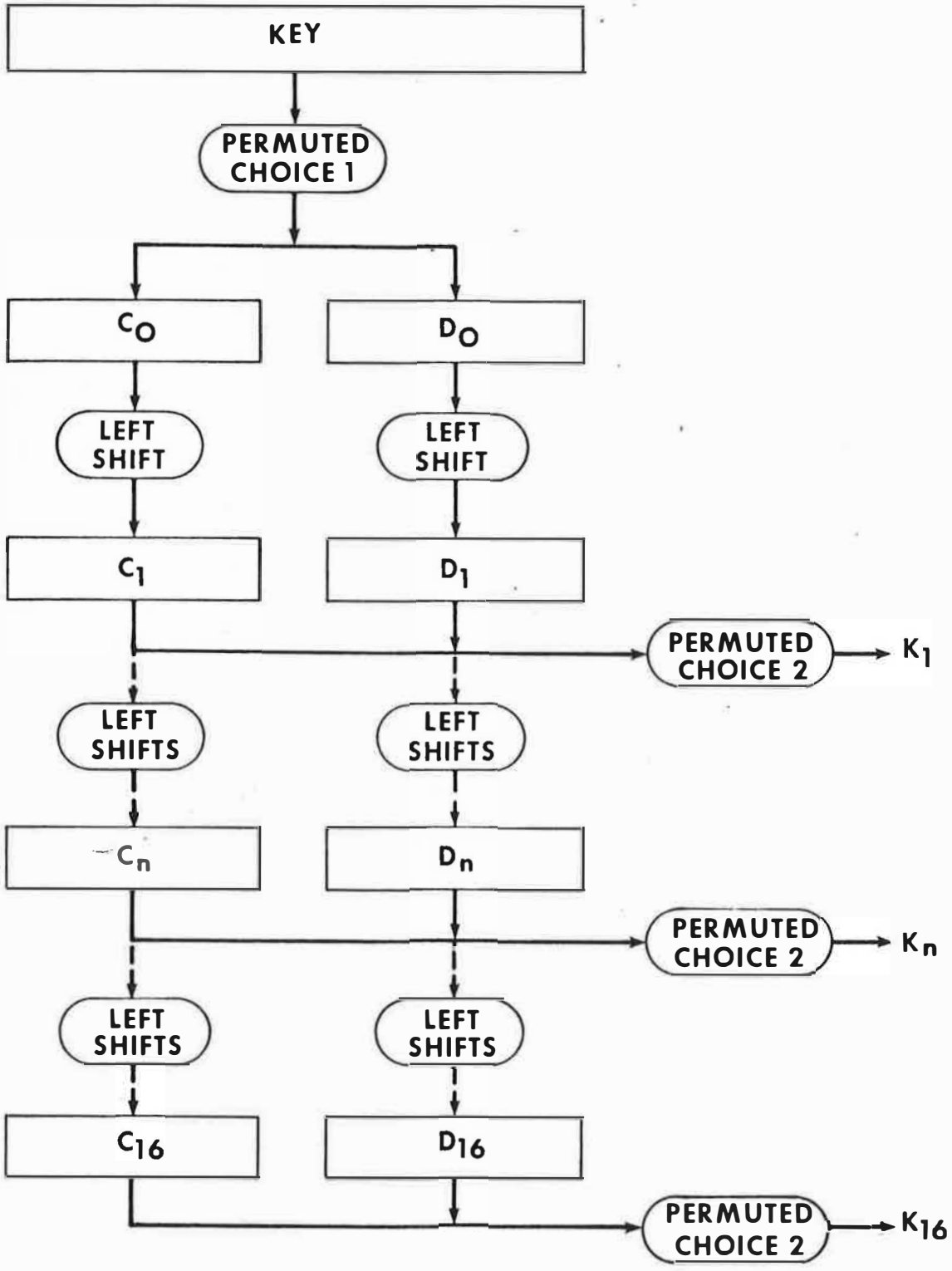


FIGURE 3. *Key schedule calculation.*

Iteration Number	Number of Left Shifts
1	1
2	1
3	2
4	2
5	2
6	2
7	2
8	2
9	1
10	2
11	2
12	2
13	2
14	2
15	2
16	1

For example, C_3 and D_3 are obtained from C_2 and D_2, respectively, by two left shifts, and C_{16} and D_{16} are obtained from C_{15} and D_{15}, respectively, by one left shift. In all cases, by a single left shift is meant a rotation of the bits one place to the left, so that after one left shift the bits in the 28 positions are the bits that were previously in positions 2, 3, . . ., 28, 1.

Permuted choice 2 is determined by the following table:

PC–2

14	17	11	24	1	5
3	28	15	6	21	10
23	19	12	4	26	8
16	7	27	20	13	2
41	52	31	37	47	55
30	40	51	45	33	48
44	49	39	56	34	53
46	42	50	36	29	32

Therefore, the first bit of K_n is the 14th bit of C_nD_n, the second bit the 17th, and so on with the 47th bit the 29th, and the 48th bit the 32nd.

NBS TECHNICAL PUBLICATIONS

PERIODICALS

JOURNAL OF RESEARCH reports National Bureau of Standards research and development in physics, mathematics, and chemistry. It is published in two sections, available separately:

• Physics and Chemistry (Section A)
Papers of interest primarily to scientists working in these fields. This section covers a broad range of physical and chemical research, with major emphasis on standards of physical measurement, fundamental constants, and properties of matter. Issued six times a year. Annual subscription: Domestic, $17.00; Foreign, $21.25.

• Mathematical Sciences (Section B)
Studies and compilations designed mainly for the mathematician and theoretical physicist. Topics in mathematical statistics, theory of experiment design, numerical analysis, theoretical physics and chemistry, logical design and programming of computers and computer systems. Short numerical tables. Issued quarterly. Annual subscription: Domestic, $9.00; Foreign, $11.25.

DIMENSIONS/NBS (formerly Technical News Bulletin)—This monthly magazine is published to inform scientists, engineers, businessmen, industry, teachers, students, and consumers of the latest advances in science and technology, with primary emphasis on the work at NBS. The magazine highlights and reviews such issues as energy research, fire protection, building technology, metric conversion, pollution abatement, health and safety, and consumer product performance. In addition, it reports the results of Bureau programs in measurement standards and techniques, properties of matter and materials, engineering standards and services, instrumentation, and automatic data processing.

Annual subscription: Domestic, $9.45; Foreign, $11.85.

NONPERIODICALS

Monographs—Major contributions to the technical literature on various subjects related to the Bureau's scientific and technical activities.

Handbooks—Recommended codes of engineering and industrial practice (including safety codes) developed in cooperation with interested industries, professional organizations, and regulatory bodies.

Special Publications—Include proceedings of conferences sponsored by NBS, NBS annual reports, and other special publications appropriate to this grouping such as wall charts, pocket cards, and bibliographies.

Applied Mathematics Series—Mathematical tables, manuals, and studies of special interest to physicists, engineers, chemists, biologists, mathematicians, computer programmers, and others engaged in scientific and technical work.

National Standard Reference Data Series—Provides quantitative data on the physical and chemical properties of materials, compiled from the world's literature and critically evaluated. Developed under a world-wide program coordinated by NBS. Program under authority of National Standard Data Act (Public Law 90-396).

NOTE: At present the principal publication outlet for these data is the Journal of Physical and Chemical Reference Data (JPCRD) published quarterly for NBS by the American Chemical Society (ACS) and the American Institute of Physics (AIP). Subscriptions, reprints, and supplements available from ACS, 1155 Sixteenth St. N.W., Wash. D. C. 20056.

Building Science Series—Disseminates technical information developed at the Bureau on building materials, components, systems, and whole structures. The series presents research results, test methods, and performance criteria related to the structural and environmental functions and the durability and safety characteristics of building elements and systems.

Technical Notes—Studies or reports which are complete in themselves but restrictive in their treatment of a subject. Analogous to monographs but not so comprehensive in scope or definitive in treatment of the subject area. Often serve as a vehicle for final reports of work performed at NBS under the sponsorship of other government agencies.

Voluntary Product Standards—Developed under procedures published by the Department of Commerce in Part 10, Title 15, of the Code of Federal Regulations. The purpose of the standards is to establish nationally recognized requirements for products, and to provide all concerned interests with a basis for common understanding of the characteristics of the products. NBS administers this program as a supplement to the activities of the private sector standardizing organizations.

Consumer Information Series—Practical information, based on NBS research and experience, covering areas of interest to the consumer. Easily understandable language and illustrations provide useful background knowledge for shopping in today's technological marketplace.

Order above *NBS publications from: Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.*

Order following *NBS publications—NBSIR's and FIPS from the National Technical Information Services, Springfield, Va. 22161.*

Federal Information Processing Standards Publications (FIPS PUBS)—Publications in this series collectively constitute the Federal Information Processing Standards Register. Register serves as the official source of information in the Federal Government regarding standards issued by NBS pursuant to the Federal Property and Administrative Services Act of 1949 as amended, Public Law 89-306 (79 Stat. 1127), and as implemented by Executive Order 11717 (38 FR 12315, dated May 11, 1973) and Part 6 of Title 15 CFR (Code of Federal Regulations).

NBS Interagency Reports (NBSIR)—A special series of interim or final reports on work performed by NBS for outside sponsors (both government and non-government). In general, initial distribution is handled by the sponsor; public distribution is by the National Technical Information Services (Springfield, Va. 22161) in paper copy or microfiche form.

BIBLIOGRAPHIC SUBSCRIPTION SERVICES

The following current-awareness and literature-survey bibliographies are issued periodically by the Bureau:

Cryogenic Data Center Current Awareness Service. A literature survey issued biweekly. Annual subscription: Domestic, $20.00; Foreign, $25.00.

Liquified Natural Gas. A literature survey issued quarterly. Annual subscription: $20.00.

Superconducting Devices and Materials. A literature survey issued quarterly. Annual subscription: $20.00. Send subscription orders and remittances for the preceding bibliographic services to National Bureau of Standards, Cryogenic Data Center (275.02) Boulder, Colorado 80302.